



**COMPETENCY STANDARD
FOR
PRODUCTIVITY IMPROVEMENT AND QUALITY CONTROL
IN FABRIC
(TEXTILE SECTOR)**

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance
Government of the People's Republic of Bangladesh

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The Competency Standard for **Productivity Improvement and Quality Control in Fabric** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry, so that individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

This document has been developed to be used in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)** to meet the skills requirements of the Textile sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades, which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The textile sector is the largest manufacturing and export-earning sector of Bangladesh, and yarn manufacturing (spinning) is the primary value-adding stage of the textile value chain. In a spinning mill, however, installed capacity alone does not determine competitiveness. Profitability and market acceptance depend on how efficiently the machines and manpower are utilized, how much of the raw material is converted into saleable yarn, and how consistently the yarn meets the quality characteristics demanded by weaving, knitting, dyeing and export buyers. A small improvement in machine efficiency, waste percentage or yarn realization, or a small reduction in yarn faults and contamination, translates directly into significant cost savings and higher value realization for the mill.

Bangladeshi spinning mills continue to report avoidable losses arising from high end breakage rates, excessive waste, unplanned downtime, contamination, count variation and customer complaints. These losses persist largely because there is a shortage of workers trained specifically to measure, analyse and improve productivity and quality using structured techniques and standard test methods. Existing training programs concentrate mainly on machine operation, while the analytical and laboratory competencies required for process control, testing, waste control, statistical process control and continuous improvement have remained largely un-standardized.

This Competency Standard has therefore been developed to prepare a competent workforce for productivity improvement and quality control in yarn manufacturing. It covers assessment of raw material quality, in-process quality control from mixing to roving, yarn testing and evaluation, identification and control of yarn faults and contamination, measurement and improvement of machine and manpower productivity, waste control and cost reduction, application of quality management systems and continuous improvement tools, and monitoring of the process environment, machine condition and instrument calibration.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and to ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competency and its corresponding Elements.

OVERVIEW:

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

1. provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
2. enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies.
3. encourages the development and delivery of flexible training that suits individual and industry requirements.
4. encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national and international process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at a training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

1. Reference to Industry Sector, Occupational Title and Occupational Description
2. Unit code
3. Unit title
4. Unit descriptor
5. Unit of Competency
6. Elements and performance criteria
7. Variables and range statement
8. Evidence guides

Together all the parts of a Unit of Competency:

1. Describe a work activity
2. Guide the assessor in determining whether the candidate is competent.

This Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competency:

1. An overview of all Units of Competency for the occupation and their corresponding duration required for completion of training.
2. The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

DESCRIPTION OF THE OCCUPATION AND THE TRAINING COURSE:

Industry Sector	Textile
Occupation Title	Productivity Improvement and Quality Control in Fabric
Occupational Description	A worker competent in Productivity Improvement and Quality Control in Fabric monitors, measures, analyses and improves the productivity and quality performance of a fabric manufacturing (weaving and knitting) mill. The occupation covers understanding of fabric types and construction (woven, knit, non-woven and braided); preparation of fabric manufacturing processes for weaving and knitting, including conventional and modern looms and warp and weft circular knitting machines; fabric quality inspection and testing in the laboratory, including appearance grading, GSM, count, EPI/PPI and CPI/WPI; identification and control of fabric defects; measurement and improvement of machine, material and manpower productivity; application of quality control tools and problem-solving techniques; implementation of continuous improvement and quality assurance practices; and monitoring of the process environment, machine condition and instrument calibration. The worker performs these tasks under general supervision, following standard operating procedures, standard test methods, control plans, machine setting sheets, safety requirements and environmental practices of the mill, and records, analyses and reports productivity and quality data using workplace formats and basic digital tools.
Level of the Course	Basic Level Skills Training
Nominal Duration	360 Hours (Approx. 03 Months)
Entry Qualification of Trainees	Grade X (SSC) passed or equivalent; candidates having a Diploma in Textile Engineering or prior working experience in a weaving or knitting mill will be given preference; physically and mentally fit to work in a fabric manufacturing mill environment; minimum age as per the prevailing labour law of Bangladesh.
Target Number of Trainees	A total of 1,500 trainees to be trained by the end of 2028
Training Providers	Textile industries and training institutes having the capacity to deliver and run the training program in accordance with this Competency Standard and the Workshop/Lab Facility Standard.
Mode of Delivery	Institution-based and/or industry-based and competency-based training combining theory, demonstration, guided practice, laboratory testing and practical performance in an actual or simulated workplace.
Employment / Job Placement	The training providers (industries and training institutes) will arrange employment for at least 65% of the total enrollees upon completion of the training.
Assessment and Certification	Competency assessment will be conducted as per the Assessment Evidence Guide of each unit, in an assessment/training centre or in an actual or simulated

	workplace, by a nationally certified assessor or occupation-specific industry expert.
Possible Job Titles	Quality Control Assistant/Inspector (Fabric), Fabric Testing Laboratory Technician, In-process Quality Checker (Weaving/Knitting), Process Control Assistant, Productivity/Work Study Assistant, Industrial Engineering Assistant, Fabric Inspection Assistant, Quality Assurance Assistant, Shift Quality Supervisor (trainee level).

UNITS & ELEMENTS AT A GLANCE:**Generic Competencies (20 Hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PIQCF-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-TEX-PIQCF-02-G	Work in a Team Environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	10
Total Duration			20 Hrs.

Sector Specific Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PIQCF-01-S	Work in the Textile Sector	1. Identify the textile sector and its main activities 2. Understand basic roles in textile production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in textile work	10
SICIP-TEX-PIQCF-02-S	Apply Green Practices in the Textile Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Duration			20 Hrs.

Occupation Specific Competencies (320 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PIQCF-01-O	Understand Fabric & Its Types	1. Identify fabric & interpret its basic characteristics 2. Describe woven fabric structure 3. Describe knit fabric structure 4. Understand non-woven & braid fabric	60 Hrs.
SICIP-TEX-PIQCF-02-O	Prepare Fabric Manufacturing Processes (Weaving and Knitting)	1. Prepare production processes for Weaving 2. Prepare production processes for Knitting 3. Demonstrate Conventional & Modern looms 4. Demonstrate Warp & Weft Circular Knitting Machines	60 Hrs.
SICIP-TEX-PIQCF-03-O	Perform Fabric Quality Inspection & Testing	1. Prepare for fabric quality inspection & testing 2. Inspect woven fabric appearance & Construction 3. Inspect Knit fabric appearance & Construction 4. Check woven fabric dimensions (GSM, Count, EPI, PPI, etc.) 5. Check Knit fabric dimensions (GSM, Count, CPI, WPI, etc.)	50 Hrs.
SICIP-TEX-PIQCF-04-O	Identify and Control Fabric Defects	1. Identify common fabric defects and their characteristics. 2. Analyze causes of fabric defects by inspection and problem-solving methods. 3. Determine corrective actions according to identified defect causes. 4. Maintain records of defects, causes and corrective actions.	50 Hrs.
SICIP-TEX-PIQCF-05-O	Improve Fabric Productivity	1. Identify factors affecting machine, labor and material. 2. Measure production output, efficiency, downtime and wastage. 3. Identify productivity losses and sources of process inefficiency.	40 Hrs.

Code	Unit of Competency	Elements of Competency	Duration (Hours)
		4. Apply appropriate methods to reduce downtime and wastage.	
SICIP-TEX-PIQCF-06-O	Apply Quality and Productivity Improvement Techniques	1. Identify quality and productivity problems. 2. Apply appropriate quality control and problem-solving techniques. 3. Implement improvement actions and approved plans. 4. Evaluate improvement results with quality and productivity targets.	30 Hrs.
SICIP-TEX-PIQCF-07-O	Implement Continuous Improvement and Quality Assurance Practices	1. Check production and quality results against set standards and targets. 2. Identify ways to improve fabric production. 3. Participate in corrective and preventive action activities. 4. Follow standard work procedures and maintain improvements 5. Record and report improvement results and suggestions.	30 Hrs.
Total Duration			320 Hrs.

Grand Total Duration of the Training Course	360 Hours (03 Months)
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COMPETENCY STANDARD: PRODUCTIVITY IMPROVEMENT AND QUALITY CONTROL IN FABRIC

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PIQCF-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1. 1 <u>OHS policies</u> and safe operating procedures are read and understood. 1. 2 Safety signs and symbols are identified and followed. 1. 3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2. 1 OHS policies and procedures are followed and practiced. 2. 2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2. 3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to the authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4. 1 Alarms and warning devices are responded to. 4. 2 <u>Emergency response plans and procedures</u> are implemented. 4. 3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables:

Variable	Range (May include but not limited to)
1. OHS policies	1. 1 International OHS requirements 1. 2 Bangladesh standards for OHS 1. 3 Bangladesh Labour Act and Rules 1. 4 Building Code 1. 5 Fire Safety Rules and Regulations 1. 6 Industry guidelines and mill safety manual
2. Personal Protective Equipment (PPE)	2. 1 Nose/face mask (dust mask) 2. 2 Ear plugs or ear muffs 2. 3 Hand gloves 2. 4 Safety shoes 2. 5 Head cap/helmet 2. 6 Apron and overalls 2. 7 Goggles and safety glasses 2. 8 Hair net
3. Hazards and risks	3.1 Physical hazards (moving machine parts, nip points, noise, vibration, heat) 3.2 Chemical hazards (lubricants, cleaning agents, solvents) 3.3 Biological hazards (bacteria, fungus in damp areas) 3.4 Ergonomic hazards (repetitive work, standing, manual lifting) 3.5 Electrical hazards 3.6 Fire hazards (cotton dust, fly accumulation, static electricity)
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid 4.5 Electrical accident response 4.6 Emergency machine stop procedures
5. First aid procedure	5. 1 Washing of open wound 5. 2 Washing chemically infected area 5. 3 Applying bandage 5. 4 Tourniquet 5. 5 Applying CPR (Cardiopulmonary Resuscitation) 5. 6 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures in a textile mill. 1.3 Emergency procedures.
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	1.3.1 Firefighting 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of hazards (biological, chemical, physical, ergonomic and electrical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 Safety signs, symbols and colour codes. 1.8 OHS awareness and reporting requirements.
2. Underpinning Skills	2. 1 Identifying OHS policies and procedures 2. 2 Following personal work safety practices 2. 3 Selecting and using appropriate PPE 2. 4 Reporting hazards and risks 2. 5 Responding to emergency procedures 2. 6 Maintaining physical well-being in the workplace 2. 7 Performing basic firefighting using fire extinguishers 2. 8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 Safety signs and symbols 4.6 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Reported incidents arising from hazards and risks to the authority. 1.4 Implemented emergency response plans and procedures. 1.5 Applied basic first aid procedure.
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2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PIQCF-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within the team and with other workers are identified.
2. Communicate and cooperate with team members	2. 1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2. 2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2. 3 Diversity in character is respected and valued in team functioning. 2. 4 Views and opinions of other team members are understood and valued. 2. 5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with the team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Support to other team members is provided to ensure the team achieves its goals. 3.4 Agreed reporting lines are followed using standard operating procedures.
4. Solve problems as a team member	4.1 Current and potential problems faced by the team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables:

Variable	Range (May include but not limited to)
1. Forms of communication	1. 1 Agenda 1. 2 Simple reports such as progress and incident reports 1. 3 Job sheets and shift handover log 1. 4 Operational manuals and SOPs 1. 5 Brochures and promotional material 1. 6 Visual and graphic materials 1. 7 Standards 1. 8 OSH information 1. 9 Signs and symbols

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Team goals and collaborative decision-making processes 1. 2 Roles and responsibilities of team members 1. 3 Relationships within the team and with other workers 1. 4 Effective interpersonal skills to interact with team members 1. 5 Effective formal and informal forms of communication 1. 6 Value of diversity in team functioning 1. 7 Correct use of workplace terminology 1. 8 Team's duties, responsibilities, authorities, objectives and task requirements 1. 9 Support mechanisms for other members of the team to ensure achievement of goals 1. 10 Methods of identifying current and potential problems faced by a team 1. 11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2. 1 Identifying team goals and collaborative decision-making processes 2. 2 Identifying roles and responsibilities of team members 2. 3 Using effective interpersonal skills to interact with team members 2. 4 Using formal and informal forms of communication 2. 5 Understanding and valuing views and opinions of other team members 2. 6 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures 2. 7 Supporting other members of the team to ensure the team achieves its goals

	2. 8 Identifying current and potential problems faced by the team 2. 9 Identifying solutions to problems 2. 10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified team goals and work processes. 1. 2 Communicated and cooperated with team members. 1. 3 Worked as a team member. 1. 4 Solved problems as a team member.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 4. Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: WORK IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PIQCF-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the textile sector. It specifically includes the tasks of identifying the textile sector and its main activities, understanding basic roles in textile production processes, recognizing common tools and equipment used, and understanding simple quality requirements in textile work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the textile sector and its main activities	1. 1 The textile sector is identified within the context of national and global industries. 1. 2 <u>Main activities</u> of the textile sector are recognized. 1. 3 <u>Different stages of the textile value chain</u> are described in simple terms. 1. 4 Key workplaces in the textile sector are identified. 1. 5 Importance of the textile sector in the economy and the environment is understood.
2. Understand basic roles in textile production processes	2. 1 <u>Basic job roles in textile production</u> are identified and described. 2. 2 Simple production workflows and task sequences are understood. 2. 3 Teamwork and coordination in production processes are identified. 2. 4 Workplace safety roles and responsibilities are understood and followed. 2. 5 Shift working practices and reporting lines are understood.
3. Recognize common tools and equipment used	3.1 <u>Common tools used in textile processing</u> are identified correctly. 3.2 <u>Basic machines and equipment used in production</u> are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance and cleaning needs of tools and equipment are recognized.
4. Understand simple quality requirements in textile work	4.1 Basic quality standards in textile work are identified. 4.2 <u>Simple defects in textile products</u> are recognized. 4.3 Importance of maintaining quality during production is understood.

Elements of Competency	Performance Criteria
	4.4 Work output is checked against basic quality expectations. 4.5 Good workmanship practices are followed to ensure quality.

Range of Variables:

Variable	Range (May include but not limited to)
1. Main activities	1. 1 Fibre production and preparation 1. 2 Yarn manufacturing (spinning) 1. 3 Fabric manufacturing (weaving and knitting) 1. 4 Wet processing (dyeing, printing and finishing) 1. 5 Garment manufacturing 1. 6 Testing and quality assurance 1. 7 Packaging, storage and export
2. Different stages of the textile value chain	2. 1 Natural and man-made fibre production 2. 2 Fibre preparation and mixing 2. 3 Yarn manufacturing (spinning) 2. 4 Fabric manufacturing (weaving/knitting) 2. 5 Dyeing, printing and finishing 2. 6 Cutting, sewing and finishing of garments 2. 7 Packaging, distribution and export
3. Basic job roles in textile production	3.1 Blow room operator/helper 3.2 Card tenter 3.3 Draw frame and comber operator 3.4 Simplex (speed frame) operator 3.5 Ring frame sider and doffer 3.6 Winding and packing operator 3.7 Quality checker/laboratory assistant 3.8 Maintenance fitter/electrician 3.9 Line leader/supervisor
4. Common tools used in textile processing	4.1 Sliver and roving cutter, scissors and nippers 4.2 Hooks, brushes and dusting tools 4.3 Spanner, screwdriver and plier set 4.4 Oil can and grease gun 4.5 Feeler and setting gauges 4.6 Wrap reel, wrap block and electronic balance 4.7 Tachometer and hand lens 4.8 Vacuum cleaner and air blowing gun
5. Basic machines and equipment used in production	5. 1 Bale opener/plucker and blow room line 5. 2 Carding machine 5. 3 Draw frame (breaker and finisher) 5. 4 Lap former and comber

Variable	Range (May include but not limited to)
	5. 5 Speed frame (simplex) 5. 6 Ring frame 5. 7 Winding machine/autoconer 5. 8 Humidification and waste collection plant 5. 9 Testing laboratory equipment
6. Simple defects in textile products	6. 1 Neps and slubs 6. 2 Thick and thin places 6. 3 Count variation and uneven yarn 6. 4 High hairiness 6. 5 Foreign fibre and contamination 6. 6 Sliver and roving breakage 6. 7 Soft/hard and damaged packages 6. 8 Oil and dirt stains

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Structure and importance of the textile sector in the national and global economy. 1. 2 Main activities and stages of the textile value chain. 1. 3 Position of yarn manufacturing within the textile value chain. 1. 4 Basic job roles, responsibilities and reporting lines in textile production. 1. 5 Simple production workflows, task sequences and shift practices. 1. 6 Common tools, machines and equipment used in textile production and their functions. 1. 7 Basic maintenance and cleaning needs of tools and equipment. 1. 8 Basic quality standards and common defects in textile products. 1. 9 Importance of quality, productivity and good workmanship in the workplace.
2. Underpinning Skills	2. 1 Identifying the textile sector and its main activities 2. 2 Describing the stages of the textile value chain in simple terms 2. 3 Understanding basic job roles and responsibilities in textile production processes 2. 4 Following simple production workflows, task sequences and teamwork practices 2. 5 Applying workplace safety roles and responsibilities in production activities 2. 6 Recognizing common tools, machines and equipment and their functions

	2. 7 Recognizing basic maintenance and cleaning needs of tools and equipment 2. 8 Recognizing simple defects and checking work output against basic quality requirements 2. 9 Following good workmanship practices to ensure consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Charts, models and samples of fibre, sliver, roving and yarn 4.3 Multimedia projector, laptop and audio-visual clips 4.4 Learning manual and handouts 4.5 Stationery

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Identified the textile sector and its main activities. 1. 2 Understood basic roles in textile production processes. 1. 3 Recognized common tools and equipment used. 1. 4 Understood simple quality requirements in textile work.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN THE TEXTILE SECTOR	Nominal Duration: 10 Hrs.	Unit Code: SICIP-TEX-PIQCF-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in the textile sector. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1. 1 Basic concepts of environmental sustainability and green practices are explained clearly. 1. 2 <u>Environmental impacts of textile production</u> processes are identified. 1. 3 <u>Resource efficiency concepts</u> are interpreted. 1. 4 Waste reduction, reuse and recycling principles are described in relation to yarn manufacturing operations. 1. 5 <u>Eco-friendly practices in the textile sector</u> are recognized and related to workplace activities.
2. Minimize resource use in the workplace	2. 1 Electricity, water, compressed air and raw materials are used efficiently according to workplace requirements. 2. 2 <u>Unnecessary consumption of resources</u> is identified and avoided during operations. 2. 3 Machines and equipment are operated in energy-efficient modes where applicable. 2. 4 Material wastage is minimized through careful handling and accurate measurement. 2. 5 <u>Resource-saving practices</u> are applied where possible. 2. 6 Leakage, spillage and wastage of materials are promptly controlled and reported. 2. 7 <u>Workplace practices</u> for resource conservation are followed.
3. Implement waste management practices	3.1 <u>Types of waste</u> generated in yarn manufacturing operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 <u>Reusable and recyclable materials</u> are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas.

Elements of Competency	Performance Criteria
	3.5 Waste disposal is carried out following environmental rules and organizational procedures. 3.6 Hazardous waste is handled carefully according to safety guidelines. 3.7 7. Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables:

Variable	Range (May include but not limited to)
1. Environmental impacts of textile production	1. 1 High electrical energy consumption of spinning machinery and humidification plants 1. 2 Cotton dust, fibre fly and micro-fibre emission 1. 3 Noise pollution from machines and compressors 1. 4 Solid waste generation (droppings, comber noil, soft and hard waste) 1. 5 Water consumption and effluent from wet processing 1. 6 Air emission and greenhouse gases from boilers and generators 1. 7 Used oil, grease and contaminated cotton waste
2. Resource efficiency concepts	2. 1 Energy-efficient motors and inverter/variable frequency drives 2. 2 Control of compressed air leakage and optimized air pressure 2. 3 Optimized humidification and ventilation control 2. 4 Reduction of soft and hard waste percentage 2. 5 Preventive maintenance to reduce power consumption 2. 6 Energy-efficient lighting and switching-off practices 2. 7 Optimum machine loading and reduction of idle running
3. Eco-friendly practices in the textile sector	3.1 Recycling of soft waste back into mixing where permitted 3.2 Use of recycled, organic or sustainably sourced fibre 3.3 Dust collection, filter and waste recovery systems 3.4 Compliance with Department of Environment (DoE) rules and regulations 3.5 Effluent treatment plant (ETP) and rainwater harvesting 3.6 Reuse of cans, bobbins, cones, cartons and packaging materials 3.7 Green building and energy audit practices
4. Unnecessary consumption of resources	4.1 Idle running of machines and motors 4.2 Compressed air leakage in pipelines and guns 4.3 Excess lighting in unoccupied areas 4.4 Over-cleaning with air guns instead of vacuum

Variable	Range (May include but not limited to)
	4.5 Excess lubricant and cleaning material use 4.6 Excessive good fibre loss with droppings and noil
5. Resource-saving practices	5.1 Switching off machines, lights and fans when not required 5.2 Reporting air, water and oil leakage promptly 5.3 Using vacuum cleaning instead of air blowing where possible 5.4 Reusing cans, tubes, cones and packaging materials 5.5 Maintaining correct machine settings to reduce waste 5.6 Recording and monitoring waste and energy use
6. Workplace practices	6.1 Standard operating procedures (SOPs) 6.2 Environmental policy of the organization 6.3 Housekeeping and 5S practices 6.4 Energy and waste monitoring records 6.5 Instructions from supervisors and safety notices
7. Types of waste	7.1 Blow room droppings and card droppings/flat strips 7.2 Comber noil 7.3 Sliver, roving and pneumafil soft waste 7.4 Hard waste from winding (yarn and thread waste) 7.5 Damaged cans, bobbins, cones and tubes 7.6 Packaging waste (poly bags, bale wrappers, straps, cartons) 7.7 Used oil, grease and oil-contaminated cotton (hazardous waste) 7.8 Used filters, sweeps and floor waste
8. Reusable and recyclable materials	8.1 Usable soft waste for back-mixing 8.2 Comber noil for coarse count and open-end spinning 8.3 Cans, bobbins, tubes and cones 8.4 Cartons, poly bags and bale straps 8.5 Metal and plastic scrap 8.6 Paper and stationery waste

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Basic concepts of environmental sustainability, green practices and climate-resilient production. 1.2 Environmental impacts of textile and yarn manufacturing operations. 1.3 Resource efficiency concepts relating to energy, water, compressed air and raw material. 1.4 Principles of waste reduction, reuse and recycling in a spinning mill. 1.5 Eco-friendly practices and technologies applied in the
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	textile sector. 1. 6 Types of waste generated at each stage of yarn manufacturing and permissible waste levels. 1. 7 Waste segregation, storage, disposal methods and compliance with environmental regulations. 1. 8 Hazardous waste handling and workplace environmental safety procedures. 1. 9 Relevant national environmental rules and organizational environmental policy.
2. Underpinning Skills	2. 1 Interpreting green concepts and relating them to workplace activities 2. 2 Identifying environmental impacts of yarn manufacturing operations 2. 3 Applying resource efficiency principles in workplace operations 2. 4 Using electricity, compressed air, water and raw materials efficiently 2. 5 Identifying and avoiding unnecessary consumption of resources 2. 6 Operating machines and equipment in energy-efficient modes where applicable 2. 7 Handling materials carefully to reduce wastage and improve accuracy 2. 8 Controlling and reporting leakage, spillage and resource wastage promptly 2. 9 Identifying and segregating different types of waste generated in yarn manufacturing 2. 10 Separating reusable and recyclable materials from non-recyclable waste 2. 11 Handling hazardous waste according to safety guidelines and regulations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedures 4. 3 Workplace documents, signs and symbols 4. 4 Codes of conduct and environmental policy

	4. 5 Waste segregation bins and storage area 4. 6 Tools, equipment and facilities appropriate to the process or activities 4. 7 Materials relevant to the proposed activity
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 Interpreted green concepts. 1. 2 Minimized resource use in the workplace. 1. 3 Implemented waste management practices.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: UNDERSTAND FABRIC AND ITS TYPES	Nominal Duration: 60 Hrs.	Unit Code: SICIP-TEX-PIQCF-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to understand fabric and its types. It specifically includes the tasks of identifying fabric and interpreting its basic characteristics, describing woven fabric structure, describing knit fabric structure, and understanding non-woven and braid fabrics.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify fabrics & interpret its basic characteristics	1.1 <u>Types of fabric</u> are identified based on manufacturing method and end use. 1.2 <u>Basic fabric characteristics</u> are described in terms of construction, weight and length. 1.3 <u>Fabric quality parameters</u> are recognized and related to end-use suitability. 1.4 Face and back of the fabric, and warp/wale and weft/course directions are identified.
2. Describe woven fabric structure	2.1 <u>Basic woven structures</u> are identified from fabric samples and diagrams. 2.2 Interlacement pattern and repeat of each basic weave are described. 2.3 <u>Woven fabric construction elements</u> are explained. 2.4 Characteristics and typical end uses of plain, twill and satin weave fabrics are described. 2.5 Derivative and fancy weaves are recognized in general terms.
3. Describe knit fabric structure	3.1 <u>Basic knit structures</u> are identified from fabric samples and diagrams. 3.2 Loop formation and stitch structure of each basic knit type are described. 3.3 <u>Knit fabric construction elements</u> are explained. 3.4 Characteristics and typical end uses of plain, rib and interlock knit fabrics are described. 3.5 Difference between weft knit and warp knit fabrics is explained.
4. Understand Non-Woven & braid fabrics	4.1 <u>Non-woven fabric types</u> and their manufacturing methods are described in general terms. 4.2 Bonding methods used in non-woven manufacturing are identified. 4.3 <u>Braided fabric structures</u> and their common applications are described.

Elements of Competency	Performance Criteria
	4.4 Differences among woven, knit, non-woven and braided fabrics are explained. 4.5 Typical end uses of non-woven and braided fabrics are identified.

Range of Variables:

Variable	Range (May include but not limited to)
1. Types of fabric	1.1 Woven fabric 1.2 Knit fabric 1.3 Non-woven fabric 1.4 Braided fabric
2. Basic fabric characteristics	2.1 GSM (grams per square metre) 2.2 Fabric width 2.3 Stitch density 2.4 Yarn count 2.5 Construction (weave/knit type)
3. Fabric quality parameters	3.1 Strength 3.2 Dimensional stability 3.3 Appearance/evenness 3.4 Hand-feel and drape
4. Basic woven structures	4.1 Plain weave 4.2 Twill weave (2/1, 3/1) 4.3 Satin/sateen weave
5. Woven fabric construction elements	5.1 Warp and weft 5.2 Selvedge 5.3 Reed count 5.4 Fabric width 5.5 Ends per inch (EPI) and picks per inch (PPI) 5.6 Crimp (%)
6. Basic knit structures	6.1 Plain 6.2 Rib 6.3 Interlock 6.4 Purl
7. Knit fabric construction elements	7.1 Wales and courses 7.2 Loop length 7.3 Stitch density 7.4 Courses per inch (CPI) and wales per inch (WPI)
8. Non-woven fabric types	8.1 Spunbond 8.2 Meltblown 8.3 Needle-punched 8.4 Thermal-bonded 8.5 Chemical-bonded
9. Braided fabric structures	9.1 Flat braid 9.2 Tubular braid

Variable	Range (May include but not limited to)
	9.3 Solid braid

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Fibre and yarn basics relevant to fabric construction 1.2 Woven fabric structures (plain, twill, satin) and their properties 1.3 Knit fabric structures (plain, rib, interlock) and their properties 1.4 Non-woven and braided fabric types and manufacturing methods 1.5 Fabric construction terminology (warp/weft, wales/courses, EPI/PPI, CPI/WPI) 1.6 Fabric quality parameters and their relationship to end use 1.7 Methods of visually and physically identifying fabric type 1.8 Difference between woven, knit, non-woven and braided fabrics 1.9 Common end uses of different fabric types
2. Underpinning Skills	2.1 Identifying fabric type from samples and swatches 2.2 Distinguishing warp/weft and wale/course directions 2.3 Recognizing plain, twill and satin weave patterns 2.4 Recognizing plain, rib, interlock and purl knit structures 2.5 Using counting glass to count threads and loops 2.6 Differentiating non-woven and braided fabric samples 2.7 Recording fabric characteristics using standard terminology
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Fabric sample board (woven, knit, non-woven, braided fabrics) 4.3 Counting glass and magnifier 4.4 Swatches showing plain, twill, satin, rib and interlock structures 4.5 Charts and diagrams of weave and knit structures 4.6 Fabric weight scale (GSM cutter and balance) 4.7 Multimedia projector, laptop and audio-visual clips 4.8 Learning manual and reference charts

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified fabric and interpreted its basic characteristics. 1.2 Described woven fabric structure. 1.3 Described knit fabric structure. 1.4 Understood non-woven and braid fabrics.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PREPARE FABRIC MANUFACTURING PROCESSES (WEAVING AND KNITTING)	Nominal Duration: 60 Hrs.	Unit Code: SICIP-TEX-PIQCF-02-O
UnitDescriptor: This unit covers the skills, knowledge and attitudes required to prepare fabric manufacturing processes. It specifically includes the tasks of preparing production processes for weaving, preparing production processes for knitting, demonstrating conventional and modern looms, and demonstrating warp and weft circular knitting machines.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare production processes for Weaving	1.1 <u>Weaving preparatory processes</u> are identified and explained. 1.2 Work order, <u>weave design/pattern</u> and warp/weft specification are obtained and interpreted. 1.3 Loom is prepared with <u>weaving preparatory materials</u> as per design. 1.4 Preparatory process parameters (warp tension, sizing add-on, count) are checked against specification.
2. Prepare production processes for Knitting	2.1 <u>Knitting preparatory processes</u> are identified and explained. 2.2 Work order, <u>knit design/pattern</u> and yarn specification are obtained and interpreted. 2.3 Knitting machine is prepared with <u>knitting preparatory materials</u> as per design. 2.4 Preparatory process parameters (yarn tension, stitch length, machine gauge) are checked against specification.
3. Demonstrate Conventional & Modern looms	3.1 <u>Types of looms</u> are identified and their working principles are described. 3.2 <u>Main parts of the loom</u> and their functions are identified. 3.3 Basic weaving motions (primary, secondary, auxiliary) are demonstrated and explained. 3.4 Differences between conventional (shuttle) looms and modern (shuttleless) looms are explained. 3.5 Loom speed, efficiency and productivity parameters are described.
4. Demonstrate Warp & Weft Circular Knitting Machines	4.1 <u>Types of circular knitting machines</u> are identified and their working principles are described. 4.2 <u>Main parts of the circular knitting machine</u> and their functions are identified. 4.3 Basic knitting motions (needle, sinker and cam action) are demonstrated and explained. 4.4 Differences between warp knitting and weft (circular)

Elements of Competency	Performance Criteria
	knitting machines are explained. 4.5 Machine speed, gauge and productivity parameters are described.

Range of Variables:

Variable	Range (May include but not limited to)
1. Weaving preparatory processes	1.1 Winding 1.2 Warping (direct/sectional) 1.3 Sizing 1.4 Drawing-in/denting 1.5 Looming
2. Weave design/pattern	2.1 Design/point paper diagram 2.2 Draft and peg plan 2.3 Denting plan 2.4 Warp and weft colour sequence
3. Weaving preparatory materials	3.1 Warp beam 3.2 Heald frame 3.3 Reed 3.4 Drop wires 3.5 Sizing ingredients
4. Knitting preparatory processes	4.1 Cone winding 4.2 Creel setting 4.3 Yarn tensioning 4.4 Machine gauge setting
5. Knit design/pattern	5.1 Stitch/loop diagram 5.2 Course/wale repeat 5.3 Yarn feeder sequence
6. Knitting preparatory materials	6.1 Cone/package 6.2 Needles 6.3 Sinkers 6.4 Cams
7. Types of looms	7.1 Conventional loom 7.2 Air-jet loom 7.3 Water-jet loom 7.4 Rapier loom 7.5 Projectile loom
8. Main parts of the loom	8.1 Weavers beam 8.2 Heald frames, heald wires and heald eyes 8.3 Reed and sley sword 8.4 Shuttle/weft insertion device 8.5 Take-up roller and let-off roller 8.6 Cloth roller
9. Types of circular knitting machines	9.1 Single and double jersey circular knitting machine 9.2 Rib and interlock circular knitting machine

Variable	Range (May include but not limited to)
	9.3 Warp knitting machine (tricot/raschel)
10. Main parts of the circular knitting machine	10.1 Needle, cylinder and dial 10.2 Cam system 10.3 Yarn feeders/creel 10.4 Take-down and cloth roller mechanism

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Weaving preparatory processes and their objectives (winding, warping, sizing, drawing-in) 1.2 Knitting preparatory processes and their objectives (winding, creel setting, machine setting) 1.3 Weave and knit design/pattern interpretation 1.4 Types, main parts and working principles of conventional and modern looms 1.5 Types, main parts and working principles of warp and weft circular knitting machines 1.6 Basic weaving motions (primary, secondary, auxiliary) 1.7 Basic knitting elements (needle, sinker and cam) 1.8 Machine settings and process parameters affecting fabric quality 1.9 Differences between shuttle and shuttle-less looms 1.10 Differences between warp knitting and weft knitting
2. Underpinning Skills	2.1 Interpreting work orders, weave/knit design and specification sheets 2.2 Setting up warp beam, healds and reed for weaving preparation 2.3 Setting creel, tension and gauge for knitting preparation 2.4 Identifying main parts and functions of conventional and modern looms 2.5 Identifying main parts and functions of warp and circular knitting machines 2.6 Observing and explaining basic weaving and knitting motions 2.7 Checking preparatory process parameters against specification
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace

4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Weaving preparatory machines (winding, warping, sizing) or models/charts 4.3 Conventional and modern (shuttle-less) loom 4.4 Warp and weft circular knitting machines 4.5 Weave and knit design/pattern samples and point paper charts 4.6 Yarn cones, warp beams, healds, reeds, needles and sinkers 4.7 Charts, models and video clips of loom and knitting machine operation 4.8 Multimedia projector, laptop and audio-visual clips 4.9 Learning manual and SOPs
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Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared production processes for weaving. 1.2 Prepared production processes for knitting. 1.3 Demonstrated conventional and modern looms. 1.4 Demonstrated warp and weft circular knitting machines.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM FABRIC QUALITY INSPECTION AND TESTING	Nominal Duration: 50 Hrs.	Unit Code: SICIP-TEX-PIQCF-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform fabric quality inspection and testing. It specifically includes the tasks of preparing for fabric quality inspection and testing, inspecting woven fabric appearance and construction, inspecting knit fabric appearance and construction, checking woven fabric dimensions (GSM, Count, EPI, PPI, etc.), and checking knit fabric dimensions (GSM, count, CPI, WPI).		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for fabric quality inspection & testing	1.1 Inspection area/equipment are selected and prepared per procedure. 1.2 <u>Fabric quality standards and specification sheet</u> are obtained and understood. 1.3 <u>Inspection and testing equipment</u> are checked for calibration and readiness. 1.4 Fabric roll/lot samples are drawn following the sampling procedure. 1.5 Abnormal conditions found during preparation are reported to the authority.
2. Inspect woven fabric appearance & Construction	2.1 Woven fabric is inspected for <u>appearance defects</u> using standard inspection method (four-point/ten-point system). 2.2 Weave structure and pattern are checked against approved sample/specification. 2.3 Fabric width, selvedge condition and edge straightness are checked. 2.4 Colour shade and pattern matching are verified against the standard. 2.5 Inspection results are graded and recorded as per <u>fabric grading system</u> .
3. Inspect Knit fabric appearance & Construction	3.1 Knit fabric is inspected for appearance defects using standard inspection method. 3.2 Knit structure/pattern and loop formation are checked against approved sample/specification. 3.3 Fabric width, edge curling and roll condition are checked. 3.4 Colour shade and pattern matching are verified against the standard. 3.5 Inspection results are graded and recorded as per fabric grading system.
4. Check woven fabric dimensions (GSM, Count, EPI, PPI, etc.)	4.1 <u>GSM (grams per square metre)</u> is measured using GSM cutter and balance. 4.2 <u>Woven Fabric count</u> is measured using counting glass.

Elements of Competency	Performance Criteria
	4.3 Fabric width and length are measured using measuring tape/scale. 4.4 Warp and weft crimp percentage are determined. 4.5 Test results are compared against specification and recorded in the prescribed format.
5. Check Knit fabric dimensions (GSM, Count, CPI, WPI, etc.)	5.1 Knit fabric GSM is measured using GSM cutter and balance. 5.2 Knit fabric count is measured using pick glass/counting glass. 5.3 Fabric width, length and loop length are measured. 5.4 Stitch density and dimensional stability (shrinkage/spirality) are checked. 5.5 Test results are compared against specification and recorded in the prescribed format.

Range of Variables:

Variable	Range (May include but not limited to)
1. Fabric quality standards and specification sheet	1.1 Buyer/technical specification sheet 1.2 Approved sample 1.3 AQL (Acceptable Quality Level) standard 1.4 In-house quality manual
2. Inspection and testing equipment	2.1 Fabric inspection machine 2.2 GSM cutter and electronic balance 2.3 Counting glass 2.4 Measuring tape/scale 2.5 Crimp tester
3. Appearance defects	3.1 Weaving defects 3.2 knitting defects
4. Fabric grading system	4.1 Four-point system 4.2 Ten-point system 4.3 Pass/fail grading against AQL
5. GSM (grams per square metre)	5.1 Standard GSM range for woven/knit fabrics 5.2 Tolerance limits
6. Woven fabric count	6.1 Ends per inch (EPI) 6.2 Picks per inch (PPI) 6.3 Reed count
7. Knit fabric count	7.1 Courses per inch (CPI) 7.2 Wales per inch (WPI) 7.3 Stitch length, stitch density and machine gauge relationship

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Fabric quality standards, specification sheets and grading systems
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	1.2 Standard fabric inspection methods (four-point, ten-point system) 1.3 Woven fabric appearance and construction defects 1.4 Knit fabric appearance and construction defects 1.5 Methods of measuring GSM, width, length and crimp 1.6 Methods of measuring EPI, PPI, CPI and WPI 1.7 Dimensional stability parameters (shrinkage, spirality) for knit fabric 1.8 Use and calibration of inspection and testing equipment 1.9 Recording and reporting of inspection and test results
2. Underpinning Skills	2.1 Preparing area and equipment for fabric inspection 2.2 Drawing fabric samples following sampling procedures 2.3 Operating fabric inspection machine and grading appearance defects 2.4 Measuring fabric GSM using GSM cutter and balance 2.5 Counting EPI/PPI and CPI/WPI using counting glass 2.6 Measuring fabric width, length and crimp percentage 2.7 Checking dimensional stability (shrinkage/spirality) of knit fabric 2.8 Comparing test results against specification and grading fabric 2.9 Recording inspection and test results in prescribed formats
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace/quality control laboratory (simulated or actual) 4.2 Fabric inspection machine 4.3 GSM cutter and electronic balance 4.4 Counting glass and magnifier 4.5 Measuring tape/scale and crimp tester 4.6 Woven and knit fabric samples with known/simulated defects 4.7 Specification sheets, AQL charts and grading formats 4.8 Multimedia projector, laptop and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Prepared for fabric quality inspection and testing. 1.2 Inspected woven fabric appearance and construction. 1.3 Inspected knit fabric appearance and construction.
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	1.4 Checked woven fabric dimensions (GSM, count, EPI, PPI). 1.5 Checked knit fabric dimensions (GSM, count, CPI, WPI).
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IDENTIFY AND CONTROL FABRIC DEFECTS	Nominal Duration: 50 Hrs.	Unit Code: SICIP-TEX-PIQCF-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify and control fabric defects. It specifically includes the tasks of identifying common fabric defects and their characteristics, analyzing causes of fabric defects by inspection and problem-solving methods, determining corrective actions according to identified defect causes, and maintaining records of defects, causes and corrective actions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify common fabric defects and their characteristics	1.1 <u>Common woven fabric defects</u> are identified and their visual characteristics are described. 1.2 <u>Common knit fabric defects</u> are identified and their visual characteristics are described. 1.3 Defects are classified according to type, location and severity. 1.4 Defect identification is recorded using standard defect terminology/codes.
2. Analyze causes of fabric defects by inspection and problem-solving methods	2.1 <u>Sources of fabric defects</u> are analyzed. 2.2 Fabric is traced back through the production process to identify probable defect origin. 2.3 <u>Problem-solving/root-cause analysis tools</u> are applied to analyze defect causes. 2.4 Findings of the analysis are discussed with the concerned section/authority.
3. Determine corrective actions according to identified defect causes	3.1 Appropriate <u>corrective actions</u> are determined based on identified defect causes. 3.2 Corrective actions are proposed to the concerned machine/process/section. 3.3 Preventive measures are suggested to avoid recurrence of the defect. 3.4 Corrective action implementation is verified through follow-up inspection.
4. Maintain records of defects, causes and corrective actions	4.1 Defects, their causes and corrective actions are recorded in the <u>prescribed defect record format</u> . 4.2 Defect data is compiled and summarized (defect frequency, type-wise analysis). 4.3 Records are reported to concerned authority as per workplace procedure. 4.4 Records are maintained/filed for future reference and trend analysis.

Range of Variables:

Variable	Range (May include but not limited to)
1. Common woven fabric defects	1.1 Slub 1.2 Thick/thin place 1.3 Broken end 1.4 Missing/float pick 1.5 Reed mark 1.6 Temple mark 1.7 Oil/stain mark 1.8 Hole/cut 1.9 Weft bar
2. Common knit fabric defects	2.1 Needle line 2.2 Drop stitch 2.3 Hole 2.4 Barre/streakiness 2.5 Oil stain 2.6 Press-off 2.7 Spirality/skewness 2.8 Pilling
3. Sources of fabric defects	3.1 Machine-related causes 3.2 Material/yarn-related causes 3.3 Process parameter-related causes 3.4 Operator-related causes
4. Problem-solving/root-cause analysis tools	4.1 Fishbone (cause-and-effect) diagram 4.2 5-Why analysis 4.3 Pareto analysis 4.4 Check sheet
5. Corrective actions	5.1 Machine adjustment/maintenance 5.2 Process parameter correction 5.3 Material replacement 5.4 Operator retraining/instruction
6. Prescribed defect record format	6.1 Daily defect log 6.2 Defect summary report 6.3 Corrective and preventive action (CAPA) record

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Common woven fabric defects and their causes 1.2 Common knit fabric defects and their causes 1.3 Root-cause analysis tools (fishbone, 5-Why, Pareto) 1.4 Relationship between machine, material, process and operator factors and defects 1.5 Corrective and preventive action principles 1.6 Defect recording and reporting formats 1.7 Basic quality control terminology and defect coding systems
2. Underpinning Skills	2.1 Identifying & classifying common woven and knit fabric defects

	2.2 Tracing fabric defects to their probable process origin 2.3 Applying fishbone diagram, 5-Why and Pareto analysis to defect data 2.4 Determining and proposing appropriate corrective actions 2.5 Suggesting preventive measures to avoid recurrence 2.6 Verifying effectiveness of corrective actions through follow-up inspection 2.7 Recording and compiling defect data in prescribed formats 2.8 Reporting defect findings and corrective actions to concerned authority
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace/quality control laboratory (simulated or actual) 4.2 Sample boards of common woven and knit fabric defects 4.3 Fabric inspection machine and magnifier 4.4 Defect record and CAPA report formats 4.5 Sample production and process flow charts 4.6 Multimedia projector, laptop and audio-visual clips 4.7 Learning manual and case study materials

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified common fabric defects and their characteristics. 1.2 Analyzed causes of fabric defects by inspection and problem-solving methods. 1.3 Determined corrective actions according to identified defect causes. 1.4 Maintained records of defects, causes and corrective actions.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IMPROVE FABRIC PRODUCTIVITY	Nominal Duration: 40 Hrs.	Unit Code: SICIP-TEX-PIQCF-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to improve fabric production productivity. It specifically includes the tasks of identifying factors affecting machine, labor and material, measuring production output, efficiency, downtime and wastage, identifying productivity losses and sources of process inefficiency, and applying appropriate methods to reduce downtime and wastage.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify factors affecting machine, labor and material	1.1 <u>Machine-related productivity factors</u> are identified. 1.2 <u>Labor-related productivity factors</u> are identified. 1.3 <u>Material-related productivity factors</u> are identified.
2. Measure production output, efficiency, downtime and wastage	2.1 Production output is measured against the planned target. 2.2 <u>Machine efficiency</u> is calculated using the standard formula. 2.3 Downtime is measured and categorized by cause. 2.4 Wastage percentage is measured and compared to the standard. 2.5 Measurement results are recorded in the prescribed production report format.
3. Identify productivity losses and sources of process inefficiency	3.1 <u>Types of productivity losses</u> are identified. 3.2 Sources of process inefficiency are analyzed using production/downtime data. 3.3 Overall Equipment Effectiveness (OEE) or equivalent indicator is interpreted. 3.4 Major loss areas are prioritized for improvement action.
4. Apply appropriate methods to reduce downtime and wastage	4.1 <u>Methods to reduce downtime</u> are applied. 4.2 <u>Methods to reduce wastage</u> are applied. 4.3 Improvement actions are implemented in coordination with the concerned section. 4.4 Results of improvement actions are monitored and recorded.

Range of Variables:

Variable	Range (May include but not limited to)
1. Machine-related productivity factors	1.1 Machine breakdown 1.2 Speed loss 1.3 Changeover/style-change time 1.4 Preventive maintenance schedule
2. Labor-related	2.1 Operator skill level

Variable	Range (May include but not limited to)
productivity factors	2.2 Absenteeism 2.3 Motivation and incentive 2.4 Manpower set-up
3. Material-related productivity factors	3.1 Raw material quality 3.2 Material availability/shortage 3.3 Material wastage
4. Machine efficiency	4.1 Standard efficiency formula (actual output/standard output × 100) 4.2 Loom/knitting machine efficiency
5. Types of productivity losses	5.1 Availability loss (breakdown, changeover) 5.2 Performance loss (speed loss, minor stoppage) 5.3 Quality loss (rejects, rework)
6. Methods to reduce downtime	6.1 Preventive/predictive maintenance 6.2 Quick style/changeover technique 6.3 Spare parts readiness 6.4 Operator training
7. Methods to reduce wastage	7.1 Process parameter control 7.2 Material handling and storage improvement 7.3 Waste monitoring and feedback system

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Factors affecting machine, labour and material productivity 1.2 Methods of measuring production output, efficiency, downtime and wastage 1.3 Concepts of availability loss, performance loss and quality loss 1.4 Overall Equipment Effectiveness (OEE) concept 1.5 Methods to reduce downtime (maintenance, quick changeover) 1.6 Methods to reduce wastage (process control, material handling) 1.7 Production reporting and data recording formats 1.8 Relationship between productivity and profitability
2. Underpinning Skills	2.1 Identifying machine, labour and material factors affecting productivity 2.2 Measuring and recording production output against target 2.3 Calculating machine efficiency 2.4 Measuring and categorizing downtime by cause 2.5 Measuring wastage percentage against standard 2.6 Identifying and prioritizing sources of productivity loss 2.7 Applying methods to reduce downtime and wastage 2.8 Monitoring and reporting results of improvement actions
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties

	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace/production floor (simulated or actual) 4.2 Production and downtime recording formats 4.3 Stopwatch/timer 4.4 Production data sheets 4.5 Efficiency and OEE calculation worksheets 4.6 Sample machine breakdown and maintenance records 4.7 Charts on productivity loss categories 4.8 Multimedia projector, laptop and audio-visual clips 4.9 Learning manual and case study materials

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified factors affecting machine, labor and material. 1.2 Measured production output, efficiency, downtime and wastage. 1.3 Identified productivity losses and sources of process inefficiency. 1.4 Applied appropriate methods to reduce downtime and wastage.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY QUALITY AND PRODUCTIVITY IMPROVEMENT TECHNIQUES	Nominal Duration: 30 Hrs.	Unit Code: SICIP-TEX-PIQCF-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply quality and productivity improvement techniques. It specifically includes the tasks of identifying quality and productivity problems, applying appropriate quality control and problem-solving techniques, implementing improvement actions and approved plans, and evaluating improvement results with quality and productivity targets.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify quality and productivity problems	1.1 <u>Quality problems</u> in the production process are identified through inspection and data review. 1.2 <u>Productivity problems</u> in the production process are identified through inspection and data review. 1.3 Problems are prioritized according to their impact and frequency.
2. Apply appropriate quality control and problem-solving techniques	2.1 <u>Quality control tools</u> are applied to analyze problems. 2.2 <u>Problem-solving methods</u> are applied to identify solutions. 2.3 Selected techniques are applied appropriately to the identified problem.
3. Implement improvement actions and approved plans	3.1 Improvement actions/plans are developed based on analysis results. 3.2 Improvement actions are approved and implemented as per workplace procedure. 3.3 Resources and responsibilities for implementation are coordinated with concerned personnel.
4. Evaluate improvement results with quality and productivity targets	4.1 Improvement results are measured against <u>quality and productivity targets</u> . 4.2 Effectiveness of implemented actions is evaluated using before/after data comparison. 4.3 Standardization of successful improvement actions is recommended. 4.4 Evaluation results are recorded and reported to the concerned authority.

Range of Variables:

Variable	Range (May include but not limited to)
1. Quality problems	1.1 High rejection/defect rate 1.2 Customer complaints 1.3 Non-conformance to specification

Variable	Range (May include but not limited to)
2. Productivity problems	2.1 Low efficiency 2.2 High downtime 2.3 High wastage 2.4 Delayed delivery
3. Quality control tools	3.1 Check sheet 3.2 Pareto chart 3.3 Histogram 3.4 Control chart 3.5 Scatter diagram
4. Problem-solving methods	4.1 PDCA (Plan-Do-Check-Act) cycle 4.2 5-Why analysis 4.3 Brainstorming 4.4 Kaizen
5. Quality and productivity targets	5.1 Defect/rejection rate target 5.2 Efficiency/OEE target 5.3 Wastage target 5.4 Delivery/output target

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Basic quality control tools (7 QC tools) 1.2 Problem-solving methods (PDCA, 5-Why, brainstorming, Kaizen) 1.3 Concept of quality and productivity targets/KPIs 1.4 Steps in implementing and evaluating improvement actions 1.5 Concept of standardization of successful improvement actions 1.6 Basic statistics for interpreting quality/productivity data 1.7 Roles and coordination required for implementing improvement plans
2. Underpinning Skills	2.1 Identifying and prioritizing quality and productivity problems 2.2 Constructing and interpreting Pareto charts, control charts and fishbone diagrams 2.3 Applying PDCA cycle and 5-Why analysis to solve problems 2.4 Facilitating/participating in brainstorming and Kaizen activities 2.5 Developing and implementing improvement action plans 2.6 Coordinating with concerned personnel for implementation 2.7 Evaluating improvement results against targets 2.8 Recommending standardization of effective improvement actions
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace/production floor (simulated or actual) 4.2 Quality and productivity data sheets/records 4.3 QC tool templates 4.4 Case studies/sample problems for practice 4.5 Improvement action plan and evaluation report formats 4.6 Multimedia projector, laptop and audio-visual clips 4.7 Learning manual and reference materials

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Identified quality and productivity problems. 1.2 Applied appropriate quality control and problem-solving techniques. 1.3 Implemented improvement actions and approved plans. 1.4 Evaluated improvement results with quality and productivity targets.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IMPLEMENT CONTINUOUS IMPROVEMENT AND QUALITY ASSURANCE PRACTICES	Nominal Duration: 30 Hrs.	Unit Code: SICIP-TEX-PIQCF-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to implement continuous improvement and quality assurance practices. It specifically includes the tasks of checking production and quality results against set standards and targets, identifying ways to improve fabric production, participating in corrective and preventive action activities, following standard work procedures and maintaining improvements, and recording and reporting improvement results and suggestions.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check production and quality results against set standards and targets	1.1 Production results are checked against <u>set production standards and targets</u> . 1.2 Quality results are checked against <u>set quality standards and targets</u> . 1.3 Deviations from standards/targets are identified and reported.
2. Identify ways to improve fabric production	2.1 Opportunities for improving fabric production are identified through observation and data analysis. 2.2 Improvement ideas are discussed with the concerned team/section. 2.3 Feasible improvement ideas are selected for implementation.
3. Participate in corrective and preventive action activities	3.1 <u>Corrective action</u> activities are participated in to resolve identified non-conformances. 3.2 <u>Preventive action</u> activities are participated in to prevent recurrence of problems. 3.3 Assigned responsibilities in CAPA activities are carried out as instructed.
4. Follow standard work procedures and maintain improvements	4.1 <u>Standard work procedures/SOPs</u> are followed consistently in daily operations. 4.2 Implemented improvements are sustained through adherence to standard procedures. 4.3 Deviations from standard procedures are reported to the concerned authority.
5. Record and report improvement results and suggestions	5.1 Improvement results are recorded in the <u>prescribed improvement/CAPA record format</u> . 5.2 Improvement suggestions are documented and submitted through the proper channel. 5.3 Records are reported to the concerned authority as per workplace procedure.

Range of Variables:

Variable	Range (May include but not limited to)
1. Set production standards and targets	1.1 Daily/shift production target 1.2 Efficiency target 1.3 Machine utilization target
2. Set quality standards and targets	2.1 Defect/rejection rate target 2.2 AQL standard 2.3 Customer specification
3. Corrective action	3.1 Immediate containment action 3.2 Root-cause elimination action
4. Preventive action	4.1 Risk identification and mitigation action 4.2 Process standardization action
5. Standard work procedures/SOPs	5.1 Machine operating SOP 5.2 Quality inspection SOP 5.3 Safety SOP
6. Prescribed improvement/CAPA record format	6.1 Kaizen/suggestion form 6.2 CAPA report 6.3 Continuous improvement log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Concept of continuous improvement (Kaizen) and quality assurance 1.2 Production and quality standards/targets and KPIs 1.3 Corrective and preventive action (CAPA) principles 1.4 Importance of standard work procedures/SOPs in sustaining improvement 1.5 Recording and reporting formats for improvement activities 1.6 Team participation and communication in continuous improvement activities 1.7 Basic quality assurance system requirements
2. Underpinning Skills	2.1 Checking production and quality results against standards/targets 2.2 Identifying deviations and reporting them to the concerned authority 2.3 Identifying and proposing ways to improve fabric production 2.4 Participating in corrective and preventive action activities 2.5 Following standard work procedures/SOPs consistently 2.6 Sustaining implemented improvements through discipline and adherence to SOPs 2.7 Recording and reporting improvement results and suggestions 2.8 Communicating effectively with team and supervisors on improvement activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties

	3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace/production floor (simulated or actual) 4.2 Production and quality standard/target sheets 4.3 SOP manuals and checklists 4.4 CAPA and continuous improvement/Kaizen suggestion forms 4.5 Sample non-conformance reports and case studies 4.6 Multimedia projector, laptop and audio-visual clips 4.7 Learning manual and reference materials

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Checked production and quality results against set standards and targets. 1.2 Identified ways to improve fabric production. 1.3 Participated in corrective and preventive action activities. 1.4 Followed standard work procedures and maintained improvements. 1.5 Recorded and reported improvement results and suggestions.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

WORKSHOP / LAB FACILITY STANDARD

Course Name:	Productivity Improvement and Quality Control in Fabric
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

1. Classroom - 400 sft (37 sqm)
2. Workshop/Lab - 1800 sft (167 sqm), OR
3. Classroom cum workshop - 2100 sft (195 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training Facilities	Required Quantity
1.	Computer/Laptop	1
2.	Multimedia projector & screen	1
3.	Printer	1
4.	Scanner	1
5.	Internet connection with Wi-Fi facility	1
6.	Whiteboard with accessories	1
7.	Cone/cheese winding machine (drum winder)	1
8.	Warping machine with creel	1
9.	Sizing machine with drying arrangement	1
10.	Shuttle loom (power loom)	1
11.	Rapier loom	1
12.	Air-jet loom	1
13.	Projectile loom	1
14.	Single/double jersey circular knitting machine	1
15.	Flat bed knitting machine (manual/semi-automatic)	2
16.	Computerized flat bed knitting machine	1
17.	Fabric inspection machine with light box	1
18.	GSM cutter and GSM balance	2 sets
19.	Electronic balance (0.001 g and 0.01 g accuracy)	2
20.	Counting glass and magnifier	10
21.	Hand tool set (spanner, screwdriver, plier, nipper)	5 sets
22.	Measuring tape and steel rule	10
23.	Fire extinguisher	2
24.	First aid box	1
25.	PPE set (dust mask, ear plug, gloves, apron, cap, safety shoes)	25 sets
26.	Stopwatch/timer (for time and motion/productivity study)	5
27.	Scientific calculator	25

The following conditions must be fulfilled –

1. The institute shall not use the same facilities for any other projects/organizations offering a similar course.
2. The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facilities required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirements of the Competency Standard (CS) provided by SICIP.

For the operation of the training course on Productivity Improvement and Quality Control in Fabric, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

Where a training institute does not possess the full-scale production machinery, the institute must establish a formal linkage with a weaving and/or knitting factory so that the practical sessions of the occupation-specific units can be conducted in an actual workplace under supervision.

The institute will also provide all other hand tools and power tools as per the CS for 25 trainees. The institute will also arrange adequate seating arrangements and classroom setup for the 25 trainees.